# Week 19: No card inside another

Adult guide built from the problem files · Unpiloted; card making and the launch unrehearsed.

This week's adult guide is week-19-facilitator.pdf, whose source is editable/facilitator-src/guide.json. These problem files hold each problem's one-sentence idea; the full keys, hints and certificates are in the guide.

## Problems and routes

| Problem | K–1 | 2–3 | 4–5 | route |
|---|---|---|---|---|
| Four small groups | P1 | | | first |
| Largest collections from the four- and eight-card decks | | | P1 | first |
| Every collection from four cards | P2 | P1 | | first |
| Search the sixteen-card deck (Grades 4-5) | | | P2 | first |
| Two-card collections from eight cards | P3 | | | first |
| The largest collections from eight cards | P4 | P2 | | first |
| Keep the starting cards (eight-card deck) | P5 | P3 | | K-1 return; 2-3 first |
| Keep the starting cards (sixteen-card deck) | | | P3 | first |
| Fewest nested rows for the four- and eight-card decks | P6 | P4 | P4 | return |
| Could four cards work? | P7 | P5 | | return |
| Set aside any one card | P8 | | | return |
| Search the sixteen-card deck (Grades 2-3) | | P6 | | return |
| Fewest nested rows for sixteen cards | | P7 | P5 | return |
| Prove the largest size from sixteen cards | | P8 | P6 | return |
| Largest collection with the circle card | | | P7 | return |
| Every largest collection | | | P8 | return |

**First route.** K–1 P1–P4; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P5–P8; Grades 2–3 P4–P8; Grades 4–5 P4–P8.

## Key, one entry per problem

### Four small groups

K–1 P1 · first route

*Idea.* Overlapping cards can share a collection; a card holding another cannot, and the empty card fits inside every card.

### Largest collections from the four- and eight-card decks

Grades 4–5 P1 · first route

*Idea.* The maxima are 2 and 3, and a cover by that many nested rows proves no larger collection works.

## Every collection from four cards

K–1 P2 · Grades 2–3 P1 · first route · wording differs by band

*Idea.* There are exactly five: each card alone and A/B; empty and AB can join no other card.

## Search the sixteen-card deck (Grades 4-5)

Grades 4–5 P2 · first route

*Idea.* The six two-symbol cards form a collection; whether seven is impossible is the question to settle.

## Two-card collections from eight cards

K–1 P3 · first route

*Idea.* Nine pairs work: two singletons, two pairs, or a singleton with the pair that lacks its symbol.

## The largest collections from eight cards

K–1 P4 · Grades 2–3 P2 · first route · wording differs by band

*Idea.* The maximum is three, reached only by A/B/C and AB/AC/BC.

## Keep the starting cards (eight-card deck)

K–1 P5 · Grades 2–3 P3 · K-1 return; 2-3 first route · wording differs by band

*Idea.* ABC and A/BC take no more cards, yet three is possible: being unable to add a card is not the same as largest.

## Keep the starting cards (sixteen-card deck)

Grades 4–5 P3 · first route

*Idea.* ABCD and A/BCD are unextendable at sizes 1 and 2, while AB/AC extends to six.

## Fewest nested rows for the four- and eight-card decks

K–1 P6 · Grades 2–3 P4 · Grades 4–5 P4 · return route · wording differs by band

*Idea.* Two and three rows suffice and are needed, since the one-symbol cards must go in different rows.

## Could four cards work?

K–1 P7 · Grades 2–3 P5 · return route · wording differs by band

*Idea.* No: three nested rows cover the deck, so any four cards put two in one row.

## Set aside any one card

K–1 P8 · return route

*Idea.* Yes: A/B/C and AB/AC/BC share no card, so removing one card leaves one of them whole.

## Search the sixteen-card deck (Grades 2-3)

Grades 2–3 P6 · return route

*Idea.* The six two-symbol cards form a collection of six; the rows that follow show it is best.

## Fewest nested rows for sixteen cards

Grades 2–3 P7 · Grades 4–5 P5 · return route · wording differs by band

*Idea.* Six nested rows cover the sixteen cards, and the six two-symbol cards need six different rows.

## Prove the largest size from sixteen cards

Grades 2–3 P8 · Grades 4–5 P6 · return route · wording differs by band

*Idea.* Six is the largest: the two-symbol cards reach it and a six-row cover allows at most one card per row.

## Largest collection with the circle card

Grades 4–5 P7 · return route

*Idea.* Four: every other card must avoid the circle, leaving a three-symbol deck whose maximum is three.

## Every largest collection

Grades 4–5 P8 · return route

*Idea.* The six two-symbol cards are the only six-card collection: singletons and, by complements, triples limit a collection to four.